INTRODUCTION

The understanding and application of probability distributions is a critical component of forecasting financial variables and asset prices. This learning module provides a foundation for understanding important concepts related to probability distributions. Regarding the application of probability distributions, this learning module explains how to construct and interpret a Monte Carlo simulation analysis. Bootstrapping, with some similarities to Monte Carlo simulations, is also demonstrated to illustrate the use and application of this statistical sampling approach.

Learning Module Overview

- The lognormal distribution is widely used for modeling the probability distribution of financial asset prices because the distribution is bounded from below by 0 as asset prices and usually describes accurately the statistical distribution properties of financial assets prices. Lognormal distribution is typically skewed to the right.
- Continuously compounded returns play a role in many asset pricing models, as well as in risk management.
- Monte Carlo simulation is widely used to estimate risk and return in investment applications. Specifically, it is commonly used to value securities with complex features, such as embedded options, where no analytic pricing formula is available
- A Monte Carlo simulation generates a large number of random samples from a specified probability distribution or a series of distributions to obtain the likelihood of a range of results.
- Bootstrapping mimics the process of performing random sampling from a population to construct the sampling distribution by treating the randomly drawn sample as if it were the population.
- Because a random sample offers a good representation of the population, bootstrapping can simulate sampling from the population by sampling from the observed sample.

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Simulation Methods

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